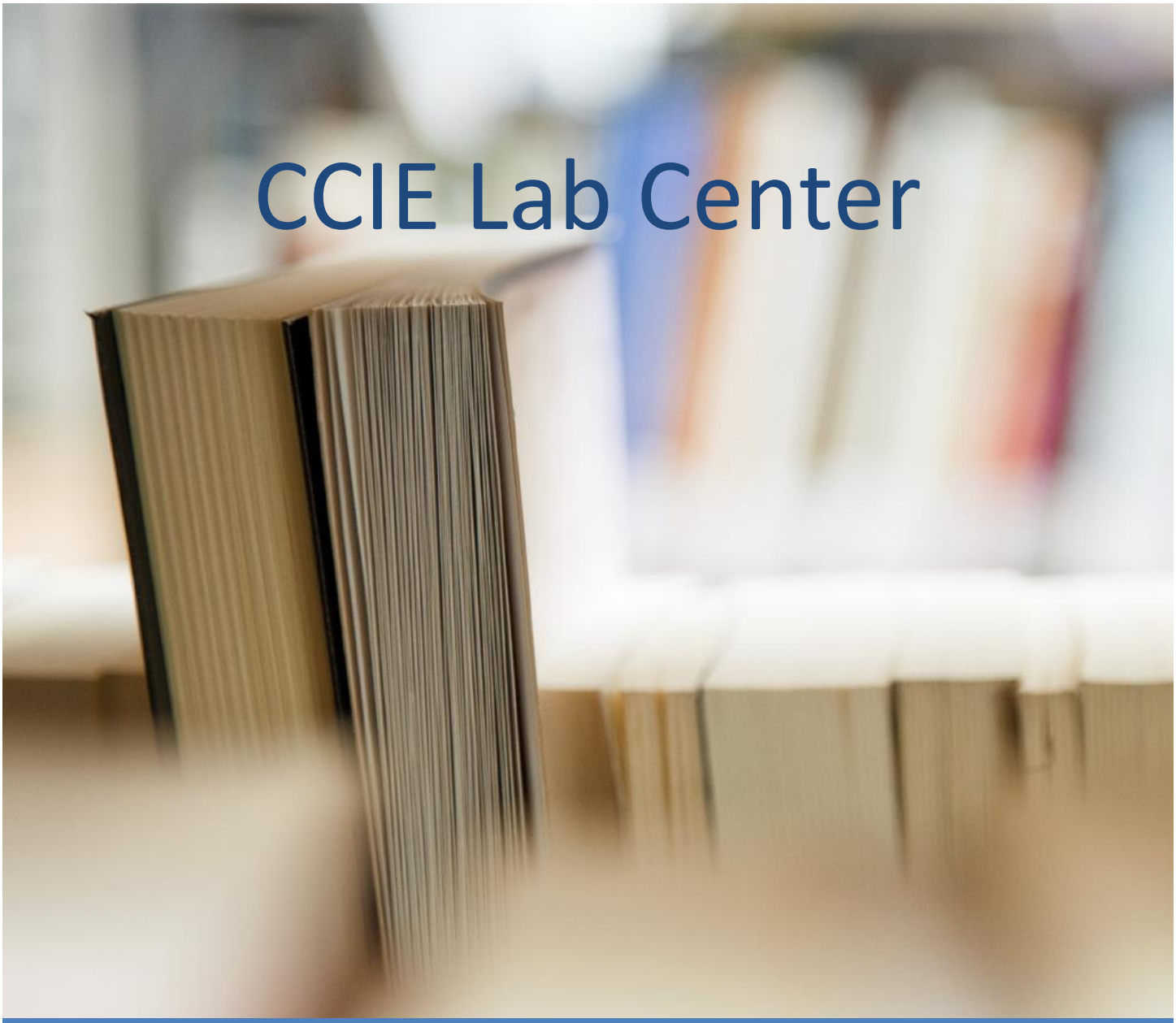
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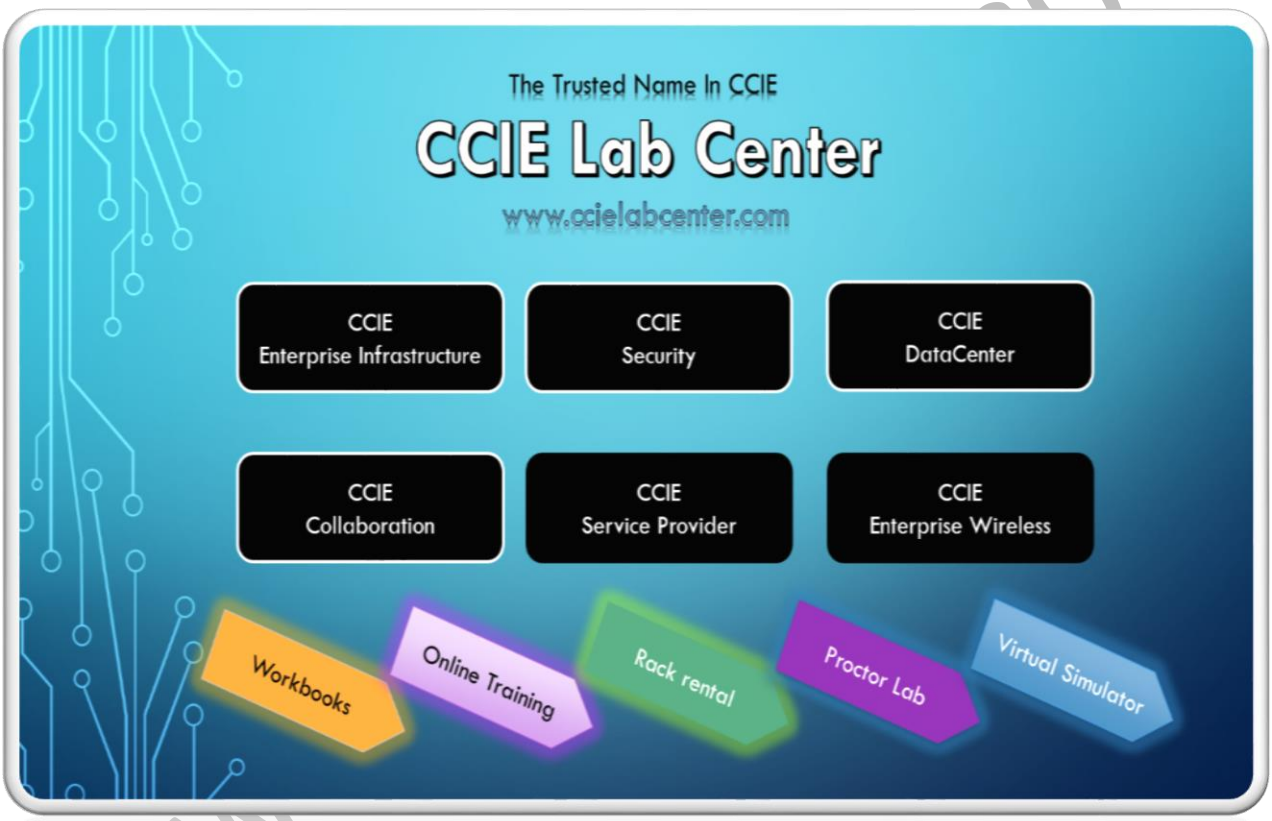


CLC - CCIE Enterprise Infrastructure v1.0: Real Lab v1.0 - Design

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CCIE Enterprise Infrastructure Real Lab v1.0-Design



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Workbook Description

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Website: https://ccielabcenter.com

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CCIE Enterprise Infrastructure Lab 1: Module 1: Design

Total 39 Questions

QUESTION 1

Welcome to the FABD2 company!

Please read all the available resources before starting the scenario by clicking 'Next item'

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QUESTION 2

Refer to the new resource(s) available.

Which action must be taken in addition to enabling Rapid PVST+ on all switches in the HQ to guarantee that the user experience is improved?

a) Disable EtherChannel Misconfiguration Guard
b) Protect ports toward end hosts with BPDU Guard
c) Configure ports toward end hosts as edge ports
d) Protect port toward end hosts with BPDU Filter

Solution

Answer: c

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Resources

RSTP Deployment

From: Network Manager;
To: CCIE Candidate;
Subject: RSTP Deployment

Hi,
We're very excited that you are with us on this journey we're about to embark on. I know we haven't even got to the official meet and greet yet, but I figured I might just take advantages of your expertise already.

Our junior engineer has been busy rolling out new switches around the HQ lately. However, it seems that STP settings have been left at their defaults on all switches. This has resulted in poor performance related to wherever a device was plugged in or removed from the network. Mangement is convinced that simply by enabling Rapid PVST+ everywhere the performance of the network will dramatically improve, but

I'm wondering if there are any additional settings that absolutely must be implemented so that, after enabling RPVST+, the user experience is truly improved.

Best regards.
Network Manager

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QUESTION 3

Refer to the new resource(s) available.

Based on the description of the issue, what is the most likely reason?

a) Rapid PVST+ requires the use of LACP fast rate to support rapid convergence on EtherChannels.
b) Trunk ports are not considered as edge ports unless explicitly configured to.
c) The MAC aging time needs to be set to a value shorter than max_age+forward_delay.
d) PortFast is not enabled globally on the switches.

Solution

Answer: b

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Resources

RE: RSTP Deployment

From: Network Manager;
To: CCIE Candidate;
Subject: RE:RSTP Deployment

Hi,

Thanks for the response. We're revisited our choices, had Rapid PVST+ enabled on HQ switches, configured ports toward end hosts with PortFast, enabled BDPU Guard on the same ports and even shortened the forward_delay_timer to 10 seconds.

Then our junior decided to test the improvement and so he configured a trunk port on sw110, connected a Linux based host there, configured it with subinterfaces for each VLAN and had a set of continuous pings running from the Linux toward both sw101 and sw102 in each VLAN while flapping the uplinks from sw110 to both distri switches. He still saw outages of around 20 seconds. What's going on there? We are considering rolling back the changes unless we can explain and fix this quickly.

Best regards
Network Manager

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QUESTION 4

Refer to the new resource(s) available.

Based on the diagram, what design change can be made to address the flapping EIGRP neighbor between r24 and r70 without impacting the network connectivity to any other DMVPN location?

a) On r70, enable EIGRP stub
b) On r21 and r70, put the WAN interfaces toward the SP into a front door VRF
c) On r70, only enable EIGRP on the r70 LAN interfaces and the DMVPN tunnel
d) On r70, do not advertise the 10.200.0.0/24 subnet in BGP
e) On r70, put the WAN interfaces toward the SP into a front door VRF

Solution

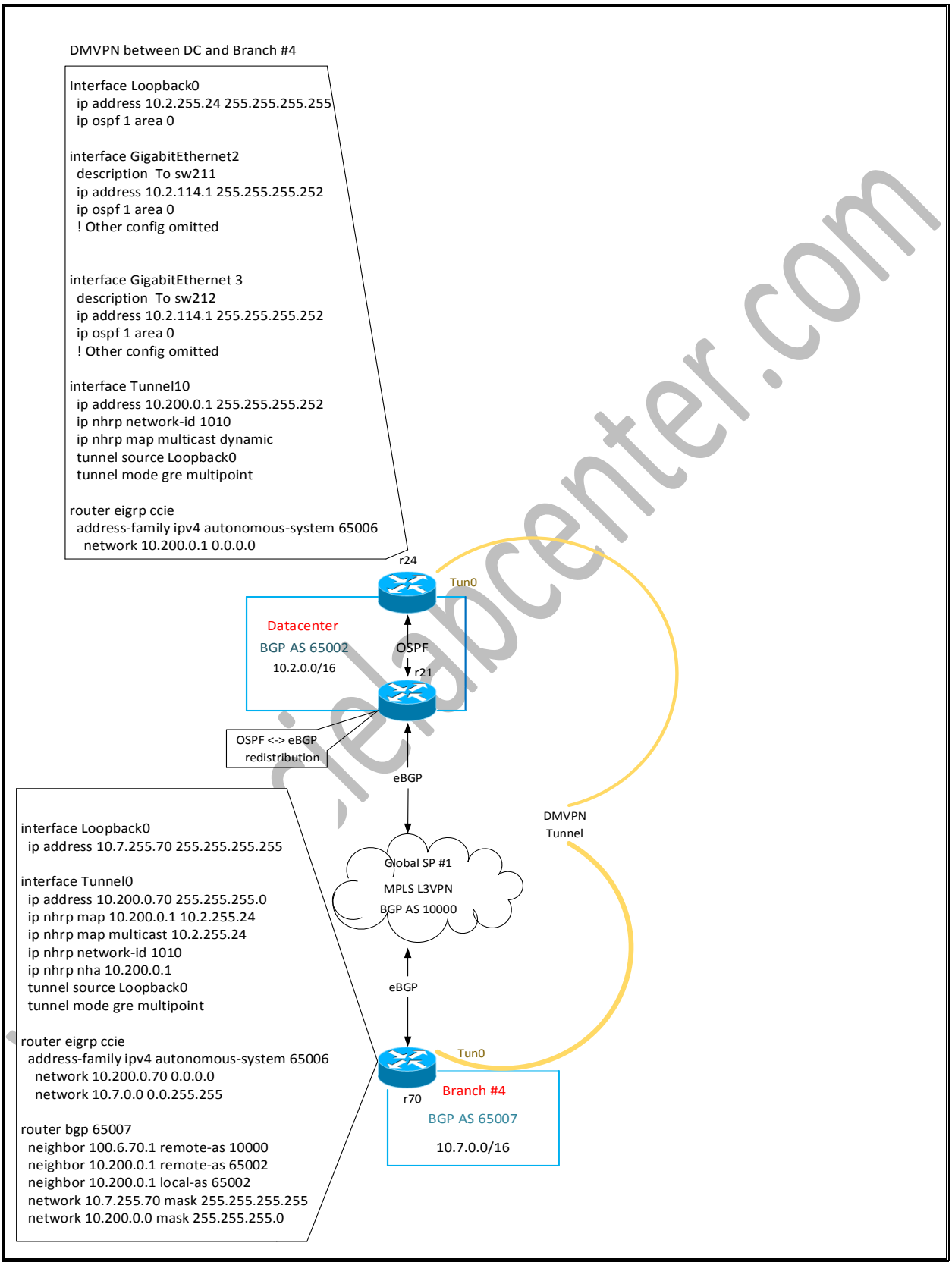
Answer:

Resources

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DMVPN between DC and Branch #4



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EIGRP Adjacency Issues - Conversation

Anna	Network Manager
Network Manager ,sorry to interrupt you while you're on those improvement at HQ but I need your help with a trouble ticket that just came in from Branch #4. They say EIGRP adjacency between r70 and r24 over the DMVPN tunnel keeps going up and down. Syslog on r70 is just filled with all kinds of logs so not sure what to focus on first. One of the things I noticed is just an ongoing churn of %DUAL-5-MERCHANGE EIGRP IPv4 65006 logs saying logs saying the EIGRP neighbor with r24 is up and then seconds later it is back down again	
	Do you know if anything changed at that branch or in those configs? isn't that branch the location where they were looking at maybe doing BGP over the DMVPN tunnel instead of EIGRP? Are any other branches having issues with EIGRP neighboring to r24 over DMVPN?
No other branches are having issues at all I just went into r24 and this is the only EIGRP neighbor that is flapping And yes, you are right. Branch #4 is the branch where they were going to try to do BGP instead of EIGRP over the DMVPN but remember, we haven't enabled r24 to do BGP over the DMVPN session with r70 anyway-I just checked it.	
	Okay… let's look at their configs and draw this all out. I am sure it is something in r70 I think remember us hitting something like this in our original deployment of the DMVPN. And let's involve our CCIE-in-making to help us with this too!

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QUESTION 5

Refer to the new resource(s) available.

For each of the EtherChannel types, indicates whether the individual statement are true, if any (select all that apply)

	Type of EtherChannel	
Statement	**LACP EtherChannel**	**Static EtherChannel**
Provides the shortest link bundling time possible		
Adds data plane overhead		
Adds control plane overhead		
Provides protection against miscabling		
Allows automatic fallback to individual link operation		
Provides the widest vendor and implementation interoperability		
Supports Layer3 EtherChannels		
Supports Layer2 EtherChannels		
Provides protection against misconfiguration		
Supports various load balancing modes		

Solution

	Type of EtherChannel	
Statement	**LACP EtherChannel**	**Static EtherChannel**
Provides the shortest link bundling time possible		
Adds data plane overhead		
Adds control plane overhead		
Provides protection against miscabling		
Allows automatic fallback to individual link operation		
Provides the widest vendor and implementation interoperability		
Supports Layer3 EtherChannels	✓	✓
Supports Layer2 EtherChannels	✓	✓
Provides protection against misconfiguration	✓	X
Supports various load balancing modes	✓	X

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Resources

NIC teaming for servers in DC

From: Huge;
To: CCIE Candidate
Subject: NIC teaming for servers in DC

Good afternoon,

My name is Huge, and I am the team lead for the sever team. Network Manager referred me to you, we had some technical details on the connectivity options for our servers and Travis suggested picking your train on this. Thanks in advance

Long story short, our severs in DC are currently connected using single uplinks to the network. We would like to migrate our servers to use NIC teaming. We're running mostly E50 and some Windows and Linux based severs and all support static and LACP-based teams –and that's the thing, We don't really understand what the pros and cons of each variants are.

I'd appreciate if you could do a comparison of the static and LACP-based NIC, teams for us. Based on this, we'll decide on how to proceed

Thanks!
Server Team

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QUESTION 6

Refer to the new resource(s) available.

What is the appropriate way to ensure that VXLAN-encapsulated traffic is properly load-balanced across physical member links of an EtherChannel, and what is the rationale to do so?

a) Use L2+L3+L4-based hash, VXLAN VTEPs randomize the source UDP port
b) Use VXLAN deep packet inspection hash, load balancing is not possible otherwise
c) Use L2+L3-based hash, VXLAN VTEPs randomize the source IP address
d) Use L2-based hash, VXLAN VTEPs randomize the source MAC address

Solution

Answer:

Resources

RE:NIC teaming for servers in DC

From: Huge;
To: CCIE Candidate;
Subject: RE:NIC teaming for servers in DC

Good afternoon,

Thank you very much for the overview of the NIC teaming choices, I think that, based on your needs, we we'll go with LACP-based traffic.

I have an additional question. We would like to achieve the maximum possible load balancing across the links in the traffic in both directions-both from and toward the servers. The majority of the traffic handled by our servers in carried in TCP and UOP, but we are also experimenting with some of the servers being VXLAN VTEPs and our concern is that because VXLAN is tunneled traffic, it will send to get polarized to just one link in the traffic. Is there a way to load-balance the VXLAN traffic too?

Thanks a lot
Server Team

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QUESTION 7

Refer to the new resource(s) available.

For each of the suggested configuration changes, indicate the event where the configuration would lead to improved convergence, if any (select all that apply)

	Event whose convergence time would be improved		
Configuration change intend to improve convergence time	Only a failure of a router or a link	Only a revival of a router or a link	Both failure and revival of a routerlink
Decrease Dead interval			
Decrease Hello timer			
Increase Dead interval			
Increase initial SPF delay			
Deploy BFD with the timer/multiplier of 100ms/3			
Increase Hello timer			
Use point-to -point network type where possible			
Decrease initial SPF delay.			

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Solution

	Event whose convergence time would be improved		
Configuration change intend to improve convergence time	Only a failure of a router or a link	Only a revival of a router or a link	Both failure and revival of a router or a link
Decrease Dead interval			
Decrease Hello timer			
Increase Dead interval			
Increase initial SPF delay			
Deploy BFD with the timer/multiplier of 100ms/3			
Increase Hello timer			
Use point-to -point network type where possible			
Decrease initial SPF delay.			

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Resources

OSPF convergence issues in DC

From: Avia;
To: CCIE Candidate;
Subject: OSPF convergence issues in DC

Hey,

We briefly met before when we were troubleshooting the flapping EIGRP adjacency between DC and Branch #4 over DMVPN. Thanks for giving us a helping hand there.

Travis said you could help with an issue we have been looking at for some time, but due to nobody here really being into the technical details of OSPF, we haven't gotten for up until now

The issue is on the convergence properties of OSPF which is currently running in our data center. Because everthing seems to work, it is running with a very basic configuration and the OSPF performance feels sloppy. What we would like to see is a sub-second convergence whenever there is a change to the network topology. Given the vast amount of settings to OSPF, we need some some pointers as to which ones are relevant for us to twisk and tune

Looking forward to your response

Regards,
FABD2 RP/WAN teams

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QUESTION 8

Refer to the new resource(s) available.

This item consists of multiple questions, you may need to scroll down to be able to see all questions.

8.1 Which two solutions for decreasing the utilization of routing tables in HQ and DC locations are applicable in FABD2's current OSPF design? (Choose two.)

a) Implementing multiple areas
b) Distribute lists
c) Summarization
d) Filter lists
e) Prefix suppression

Solution

Answer:

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8.2 For every solution intended to control the utilization of the routing tables in FABD2 HQ and DC, select the correct characteristics if any. (select all characteristics that apply.)

	Characteristics		
Solution	Controls the distribution scope of Type-1/Type-2 LSAs	In most cases, config-and-forget	In most cases, requires ongoing operational maintance
Distribute lists			
Implementing multiple areas			
Summarization			
Prefix suppression			
Filter lists			

Solution

	Characteristics		
Solution	Controls the distribution scope of Type-1/Type-2 LSAs	In most cases, config-and-forget	In most cases, requires ongoing operational maintance
Distribute lists			
Implementing multiple areas			
Summarization			
Prefix suppression			
Filter lists			

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8.3 What are the two disadvantages of using distribute list to control the routing table contents in FABD2 HQ and DC? (Choose two.)

a) Incorrect deployment of distribute lists may cause permanent routing loops
b) OSPF link state database contents may become inconsistent
c) SPF algorithm will need more time to complete due to examining LSA contents against the distribute list
d) Distribute links in OSPF have no influence on the contents of the CEF FIB on the router
e) Administrative overhead will grow since distribute lists must be deployed on all OSPF routers

Solution

Answer:

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Resources

RE: OSPF convergence issues in DC

From: Avia;
To: CCIE Candidate
Subject: RE: OSPF convergence issues in DC

Hey,

We liked those ideas about OSPF you brought in!

Now that we're on the topic, there's another thing bugging me. You see we have our HQ and DC and redundantly connected through sw101/sw201 and sw102/sw202 links and since everything is put into OSPF was 0, the inputting tables in HQ and DC are quite a mess-swamped with routes

Is the just we have been using inbound distributes lists to control the size of the routing tables. In your opinion. could that still be relevance?

What other options do we have? Sure, we can re-engineer out network and start splitting it into areas but that's too intrusive it's something we'd like to avoid possible. One thing. The topology of our infrastructure is HQ and DC is very unlikely to change, but end host networks (stub networks) can come and go quite frequently including being submitted or lumped back together.

Thanks!
FABD2 RP/WAN teams

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QUESTION 9

Refer to the new resource(s) available.

This item consists of multiple questions, You may need to scroll down to be able to see all questions.

9.1 Based on current FABD2 design, which switch or switches must perform DHCP Snooping to avoid DHCP-related incidents in the HQ?

a) sw110 and sw211
b) sw110
c) sw101, sw102, sw110 and sw211
d) sw101, sw102 and sw110

Solution

Answer:

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9.2 If DHCP Snooping was activated on sw110, what interfaces would need to operate as trusted interfaces?

a) Port channels toward sw101 and sw102
b) SVI for management VLAN on sw110
c) SVIs for VLANs where DHCP Snooping is activated
d) Ports toward end hosts

Solution

Answer: a

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9.3 Which of the following two approaches can be used to avoid breaking DHCP functionally when the DHCP server runs on a different device than the DHCP snooping device? (Choose two)

a) On IOS based DHCP servers and relay agents, accept DHCP messages containing Option 82 having all-zero giaddr
b) On switches performing DHCP Snooping, disable Option 82 insertion
c) On DHCP servers, allocate IP addresses to clients based on Option 82 remote-id and circuit-id values instead of client MAC addresses
d) On DHCP clients, preconfigure customized Option 82 contents
e) On IOS-based DHCP relay agents, change the relay policy to replace Option 82

Solution

Answer:

Resources

Improving DHCP security in HQ

From: Network Manager;
To: CCIE Candidate
Subject: Improving DHCP security in HQ

Hello,

In HQ, we recently had an issue when an employee came in, plugged in his laptop and he forgot he had a linux server VM running in his VirtualBox with a DHCP server enabled. You can imagine the mess we've faced and it took us ages to even find out where the offending DHCP server is!

We understand that the DHCP Snooping feature should be a reasonable protection against it however, when searching for more details, we ended up with conflicting recommendations. So can you clarify a couple of questions for us? Just as an reminder, all current and future end hosts on HQ will be connected only to sw110. Our sw101 and sw102 are DHCP Relay/Agents toward the DHCP server located on sw211

Thanks! Oh and I appreciate you helping out Avia and Huge

Network Manager

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QUESTION 10

Refer to the new resource(s) available.

What are two parallel reasons for the direct spoke-to-spoke DMVPN tunnel coming up between r62 and r70? (Choose two)

a) Shortcut switching is enabled on the DMVPN tunnel of r62 and r70
b) The EIGRP next-hop self feature is disabled on r24
c) NHRP Redirects are enabled on the DMVPN tunnel of r24
d) r62's NHRP and r70's NHRP registrations can be seen by each other as they are multicasted over the same DMVPN tunnel
e) Shortcut switching is enabled on the DMVPN tunnel of r24
f) NHRP Redirects are enabled on the DMVPN tunnel of r62 and r70

Solution

Answer:

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Resources

Unexpected tunnel between r62 and r70

Avia	Travis
Travis, sorry to bother you again with this but something would is going on Actually. I think it has been going on for a while now, but we only noticed it while we were troubleshooting that EIGRP neighbor issue between r24 and r70. What we saw was that in r70 there was a DMVPN tunnel up with Branch #3 r62 router. r70 even learned a network connected to r62 even though this should not be possible since our hub summarizes everthing to the default route when speaking to spokes. Our NOC book for troubleshooting DMVPN tunnels at the branches does not list this as something we should see. So cliently this didn't use to be case in the past when we made that book. What I suspect is this. Do you remember last year when we hired that consultant to recommend changes to the DMVPN networks? I think it is possible that during the design review we must have missed something	
	Wow, that's been a while ago I don't think we still have the document of his exact designs and config recommendations. I know we just cleared all of them in the designs review and put them in the network. Hmm…. Have you asked our CCIE apprentice about his opinion yet?

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QUESTION 11

Refer to the new resource(s) available.

Based on the requirements for the security hardening in Branch #3, what is a viable solution?

a) Protected ports
b) VLAN ACLs
c) Private VLANs with two independent community secondary VLANs
d) Private VLANs with an isolated secondary VLAN
e) Port ACLs
f) Private VLANs with an isolated and a community secondary VLAN

Solution

Answer:

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Resources

Tightening down security in branch #3

From: 'Travis Handerson'<.com>
To: 'CCIE candidaye'<.com>
Subject: Tightening down security in branch #3

Hello,

Thank you very much for classifying that tunnel between Branches #3 and #4. I really got tense about that DMVPN – seems like it keeps coming in with mysteries Anyway speaking about Branch #3. I realized I wanted to ask you something

At Branch #3 we have been considering improving the security of the end hosts. We have several public-operated terminals connecting there that would best be kept seprated from each other and also from anyone else. There is also a series of less-than-perfectly trusted end hosts that need to talk to each other, but again it would be bother if they, as a group. didn't have access to anyone else on the branch

I don't want to spend my IP addressing space on these devices excessively identify all the terminals and the end hosts I just mentioned should be kept in a single IP network. While we currently have only a single access layer switch on the site are may be extending that in the future or –if the budget is too tight –reuse some of the free ports on the distribution layer switches for the purpose of connecting these hosts. We'd like to solve this with an-approach that's easy to maintain works community across multiple switches and if possible. Does not require us to go too crazy with ACLs defining who can talk to whom. Can you support a solution

Thanks – much appreciated

Travis
Network Manager

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QUESTION 12

Refer to the new resource(s) available.

Drag the QoS configuration action on the left to the correct device on the right, observing the correct order of the configuration. Not all options are used

	r24	r70
Create parent QoS policy with 10Mbps shaper	1st Action	Action
Create parent QoS policy handling traffic classes	2nd Action	
Create child QoS policy Handling traffic classes	3rd Action	
Apply the child QoS policy as an NHRP-mapped policy on the tunnel	4th Action	
Configure the NHRP QoS group name		
Apply the parent QoS policy as a service policy on the tunnel		
Associate the child QoS policy with the parent QoS policy		
Apply the parent QoS policy as an NHRP-mapped policy on the tunnel		
Create child QoS policy with 10Mbps		

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Solution

	r24	r70

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Resources

Slowness in Branch #4

From: 'Avona Vany;l'<.com>
To: 'CCIE candidaye'<.com>
Subject: Slowness in Branch #4

Hey,

Wondering if you could help us with something.

We have found that whenever Branch #4 complained about having this issue, 100% coincided with peak volumes of traffic coming in on r70 interface towards the MPLS service provider. So clearly, we need to fix this, but we aren't sure what to do or where to do it. We had choices, but we really don't want to impact any of the other branches, so we thought we'd come to you. Really hope you can help us.

We aren't sure what information you might need, so here is everything we could think of

- Qos is not configure on any of the devices in the datacenter
- r21 has a 1Gbps connection with the SP, with a contracted rate of 1Gbps and 4 traffic classes
- r70 has a 1Gbps connection with the SP, with a contracted rate of 10Mbps and 4 traffic classes

If you could help us figure out how to address the issue that would be great

Avona
FABD2 RP/WAN team

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QUESTION 13

Refer to the new resource(s) available.

What change is required to the BGP configuration in the environment of Global SP #1 so that r4 learns about multiple paths to networks at Branch #3?

a) On r5 and r6, activate the route reflector function
b) On r5 and r6, unique RDs need to be configured
c) On r3 as the route reflector, BGP Multipath feature must be enabled
d) On each PE, unique RTs need to be configured
e) On r4 the BGP maximum paths setting needs to be increased

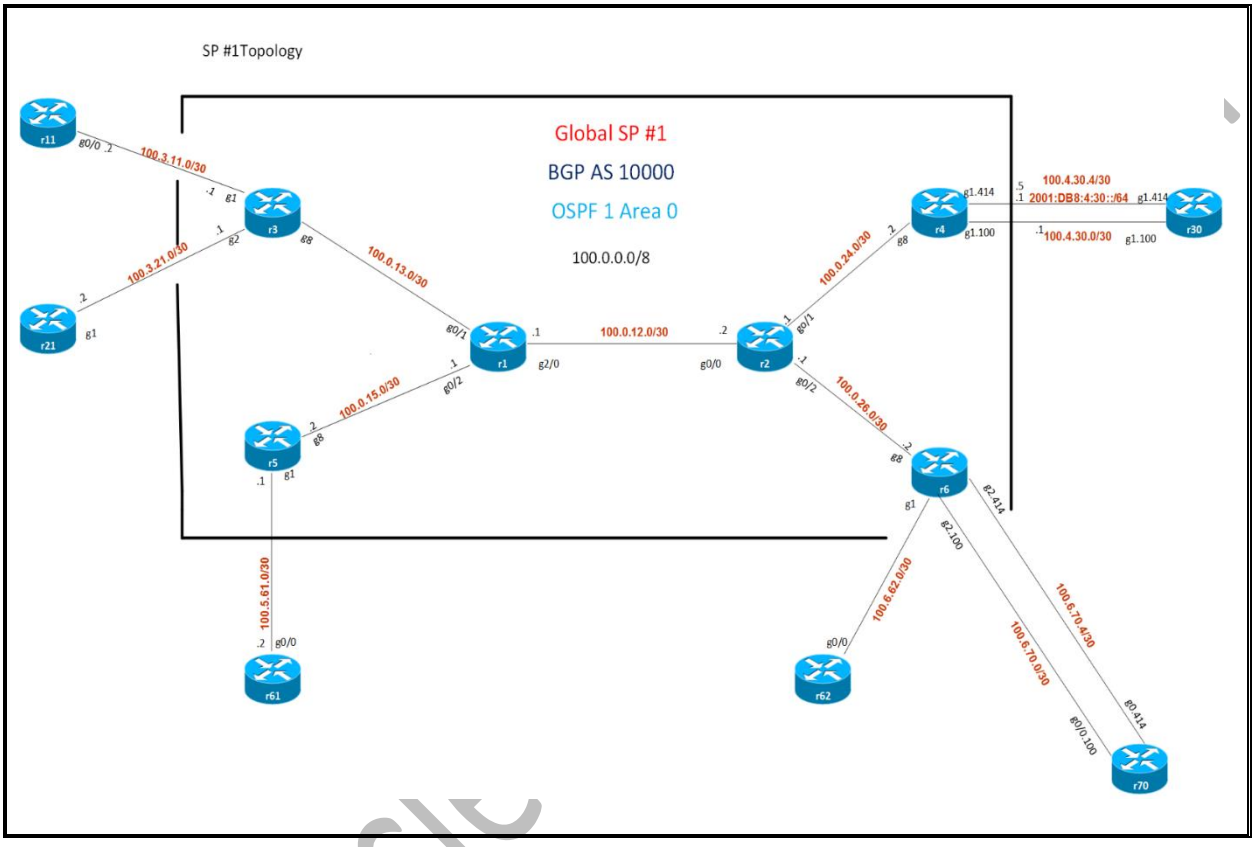
Solution

Answer:

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Resources



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FW: Regarding BGP routes at our SP

From: 'Avona Vany;l'<.com>
To: 'CCIE candidaye'<.com>
Subject: FW: Regarding BGP routes at our SP

Hello,

I noticed something strange the other day in our SP's network. I was in r4 and I looked at the routing table for the FABD2 VRF and saw the Branch #3 LAN prefix listed in the BGP table only once I was surprised because I thought I would see if twice. But no, only listed once.

Travis took a quick look at the SP design and implementation specs and said he thought he knew what it likely was, but he was just leaving for vacation for a week, so he sent me to you. He said you'd be able to figure out. He sent me some quick notes, that I have added at the bottom of the email.

Avona
FABD2 RP/WAN team.

From: 'Travis Handerson'<.com>
To: 'Avona Vanyl'<.com>
Subject: Tightening down security in branch #3

Avona,

Sorry I can't be of more assistance right now. just about to go offline for one week of hiking in the mountain.

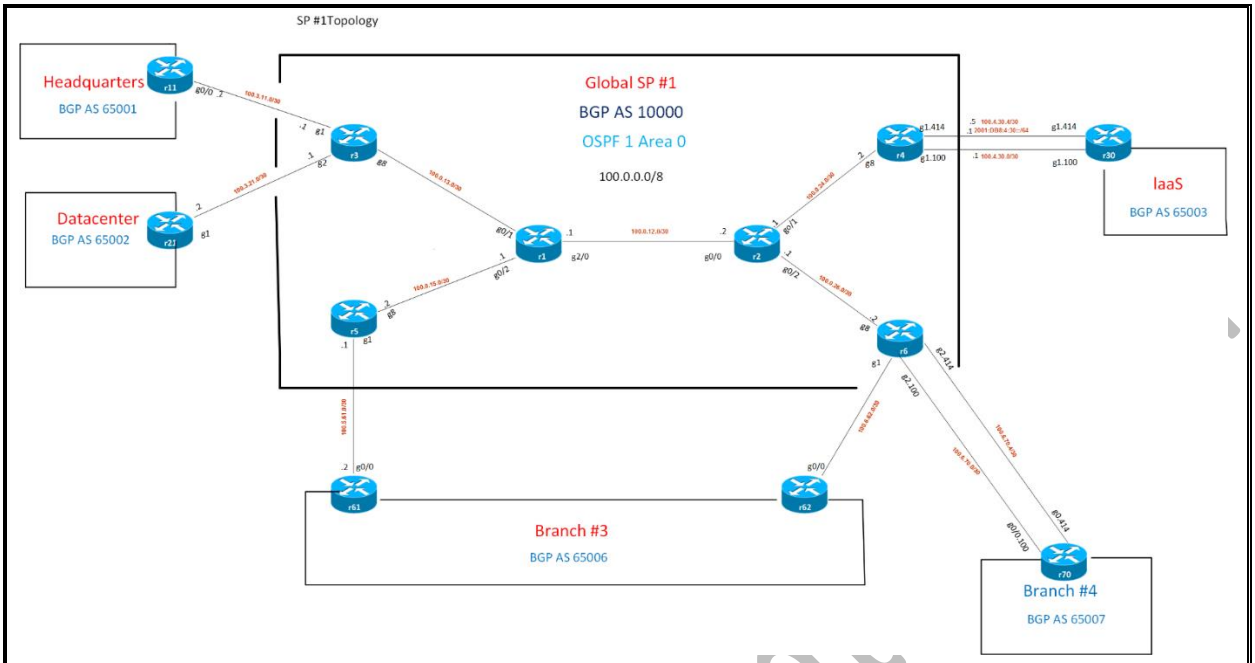
I had a quick look at the SP side of things and this is the overall design when it refuses to us (short and concise – writing this on my cellular)

- At our location are place into VRF "fabd2" on relevant PEs (r3,r4,r5 and r6)
- RT 100000 1 for import and export
- RD 100000 1 on all PEs
- LDP deployed on Ps and PEs to distribute labels
- BGP used to advertise our prefixes and associated VPN tables
- r2 is the BGP route reflector, r4, r5 and r6 have BGP sessions only to r3 as its RR clients

I happened to have a network topology diagram in my phone. Here's the relevant part.

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I'm sure our CCIE prospect can assist you in figuring out what the problem is.

Travis

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QUESTION 14

Refer to the new resource(s) available.

Which two addresses are the best choices for the Connected FABD2 and RapidStreaming multicast groups? (Choose two.)

a) 232.2.1.1
b) 232.1.1.1
c) 239.129.1.2
d) 239.2.1.1
e) 232.129.1.1
f) 239.1.1.2
g) 239.1.1.1

Solution

Answer:

Resources

Corporate multicast

From: 'Travis Handerson'<.com>
To: 'CCIE candidaye'<.com>
Subject: Corporate multicast

Hi,

Just got of the monthly management meeting and as seen so other before, this gives us some work to do.

Following the concerns of our branch-based employee about feeling disconnected from what is going on at the HQ. Our HR conceived the idea of an internal 'TV channel' led to all our remote branches source from the HQ. They still hasn't decided exactly what kind of content that will be in the feed, but they have decided on the name ConnectedFABD2

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Now, this won't be the first multicast deployment in our network. We already have a multicast feed deployed in HQ named JustStreaming. These are some quick facts about it.

JustStreaming

- Already existing in HQ
- High bandwidth
- User group 239.1.1.2
- Sources and receivers only in HQ

Regarding the ConnectedABD2 feed properties, we assume the following

ConnectedFABD2:

- New feed to be introduced
- Low bandwidth feed, at least initially
- Sources will be located in HQ
- Receiving spread across all DMVPN based branches as well as in HQ

Finally….the JustStreaming feed was, in this time deployed in a somewhat rushed and 'just-make-it-work' manner. We are looking at eventually replacing it with another feed we call RapidStreaming which we would this to design properly this time. Assumed properties of RapidStreaming are

RapidStreaming

- New feed to be introduced
- High bandwidth feed
- Will eventually replace JustStreaming
- May need to co-exist with JustStreaming for a while
- Sources and recievers only in HQ

With all of this in mind, I'd like you to go over things and provide me with some assistance on hour to approach this. In addition to what I have already started above, please also consider the following points.

- Access and evaluate the IP multicast addressing space in use, do we continue to use it or should we find something else?
- Placement of rendezvous points(RPs) for Connected FABD2 as well as RapidStreaming . Since both feeds are of high importance, it is essential that we have a resilient RP setup. Whether it's active or primary backup is not important to me. I'll listen to your suggestions.
- RP address advertisement across the network, should we use any dynamic RP advertisement mechanism, or should we go with static RP address configuration

Thanks in advance
Travis
FABD2 Network Manager

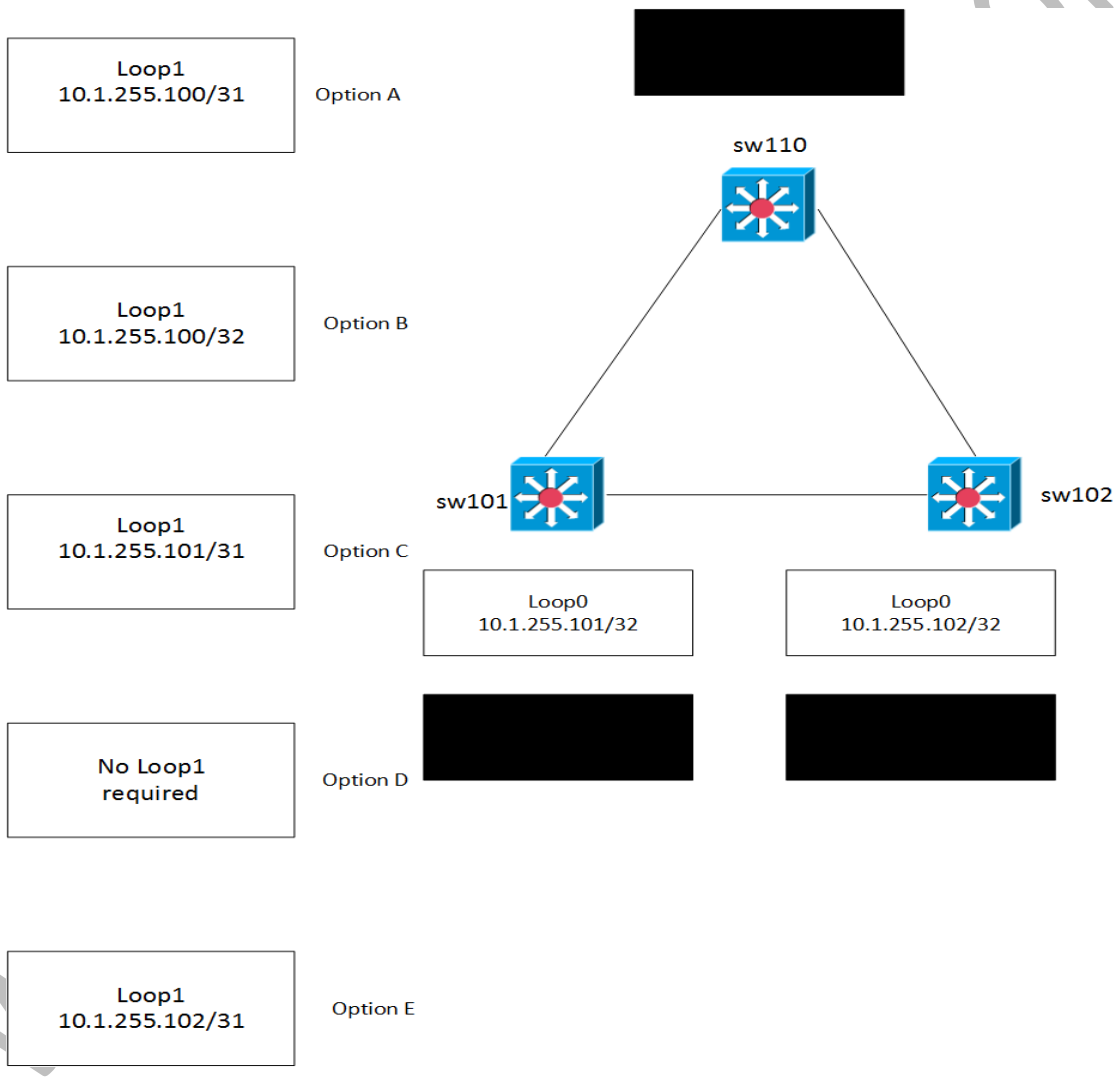
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QUESTION 15

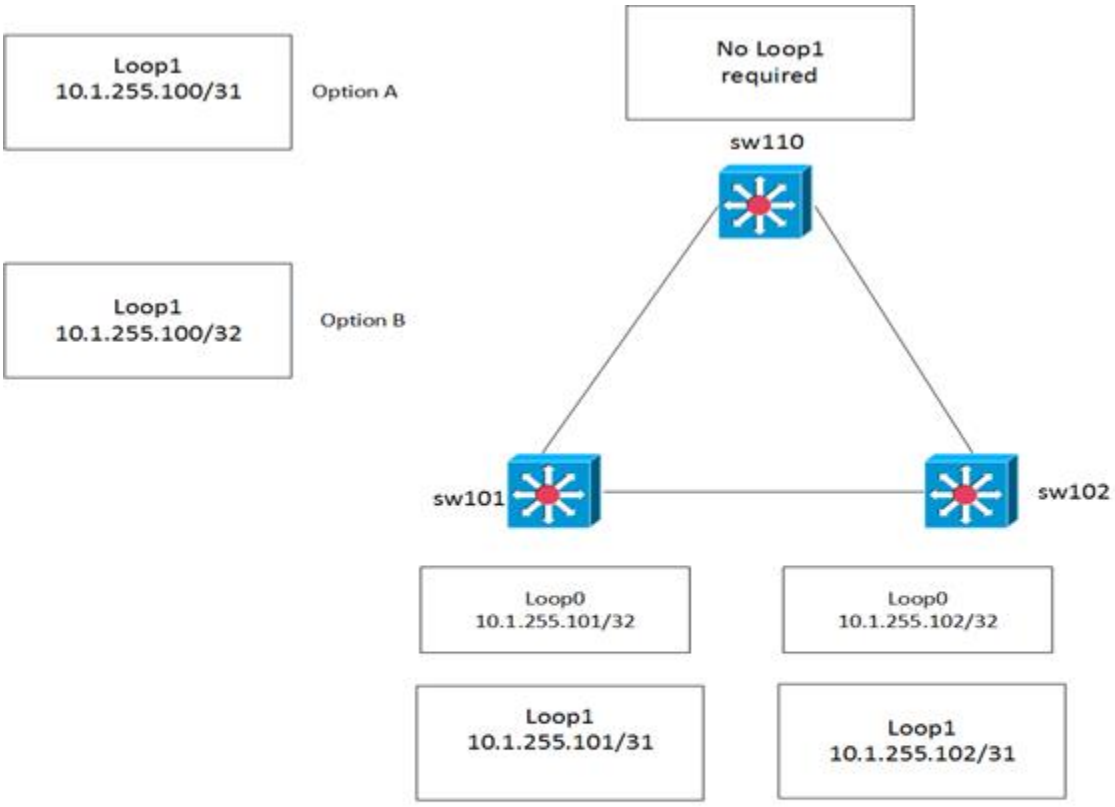
Refer to the new resource(s) available.

Considering the intended RP design for the High Bandwith multicast range, drag and drop the appropriate Loop1 configuration on the left to each switch in the diagram. Any Loop1 configuration can be dropped to multiple switches. Not all options are used



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Solution



CCIE Lab Center

Resources

RE: Corporate multicast

From: 'Travis Handerson'<.com>
To: 'CCIE candidaye'<.com>
Subject: RE: Corporate multicast

Hi,

Thanks for your input. Based on that and trying to cable in all other comments I received , I have made the following decisions for our multicast designs.

For **High Bandwidth** multicast feeds (currently JustStreaming and RapidStreaming)

- Allocated multicast group range 232.1.1.1 thru 239.1.255.255
- sw101 and sw102 will act as RPs in a high-availability RP design using the IP address 10.1.255.100 as the RP for the entire High Bandwidth multicast range
- sw110 is a layer2 only switch and will only run IGMP Snooping and perhaps P/M Snooping if it is supported there (no big issue if it isn't)
- On routers and multilayer switches, the RP address will be configured statically
- 239.1.1.101 will be used for RapidStreaming

For **Low Bandwidth** multicast feeds (currently ConnectedFABD2)

- Allocated multicast group range 239.201.1.1 thru 239.201.255.255
- Any legacy branch may have receivers that are eligible to join any Low Bandwidth multicast group
- 239.201.1.101 will be used for ConnectedFABD2

Please keep this in intend when designing the multicast solution for us

Travis
FABD2 Network Manager

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QUESTION 16

Refer to the new resource(s) available.

Considering correct FABD2 design, which two devices are the best choices for placement of the RP for Low Bandwidth multicast streams? (Choose two.)

a) sw101
b) r11
c) sw102
d) r21

Solution

Answer:

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CCIE Lab Center

Resources

RE: RE: Corporate multicast

From: 'Travis Handerson'<.com>
To: 'CCIE candidaye'<.com>
Subject: RE :RE: Corporate multicast

Hey,

OK, so I apparently opened a can of worms, when I asked my teams about RP placement for our Low Bandwidth multicast ranges.

Everybody agrees that we will be using redundant RPs with MSDP, but this is where agreement stops. Everyone seems to have strong opinions on where the RP role should be placed in the network. Below is a messages of the group that we had earlier today (I have redacted the names to 'protect' the professional reputation of my team members 0)

- Team member #1: "Let's just use sw101 and sw102 also as the RPs for all the Low Bandwidth multicast address stream range."
- Team member #2: "I don't know if that is wise since those are also the RPs for the High Bandwidth range"
- Team member #3: "Since the sources for all the low bandwidth traffic will be in HQ is VLANs handled by sw101 and sw102, RP placement these is really the best "
- Team member #4: "Well, since receivers in all DMVPN branches will be able to request a low bandwidth stream, why not put the RPs of the WAN edge, more specially r11 and r21, as they are both directly connected to the MPLS cloud? Placing the RPs there is the connect designs as they will be replicating the multicast traffic from the LAN out the MPLS WAN anyway"

I'm not sure which direction to take from here. All soon like valid ports to the. What do you recommend?

Travis
FABD2 Network Manager

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QUESTION 17

Refer to the new resource(s) available.

What prefixes, along with their label bindings must be advertised by LDP in the MPLS mock lab to enable MPLS L3VPN services?

a) Loopback0 prefixes of all PE routers and prefixes of all infrastructure links
b) Loopback0 prefixes of all PE and P Routers
c) Loopback0 prefixes of all PE routers
d) Loopback0 prefixes of all PE and p routers, and prefixes of all infrastructure links

Solution

Answer:

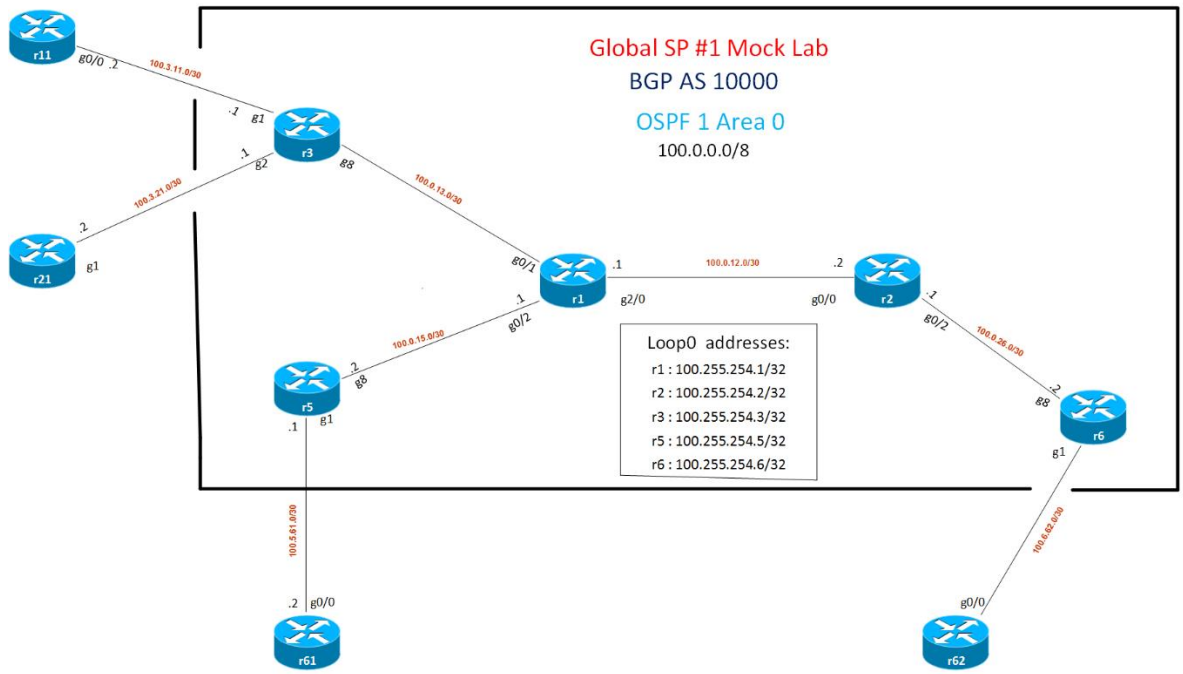
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Resources

SP #1 Mock Lab

SP #1Topology



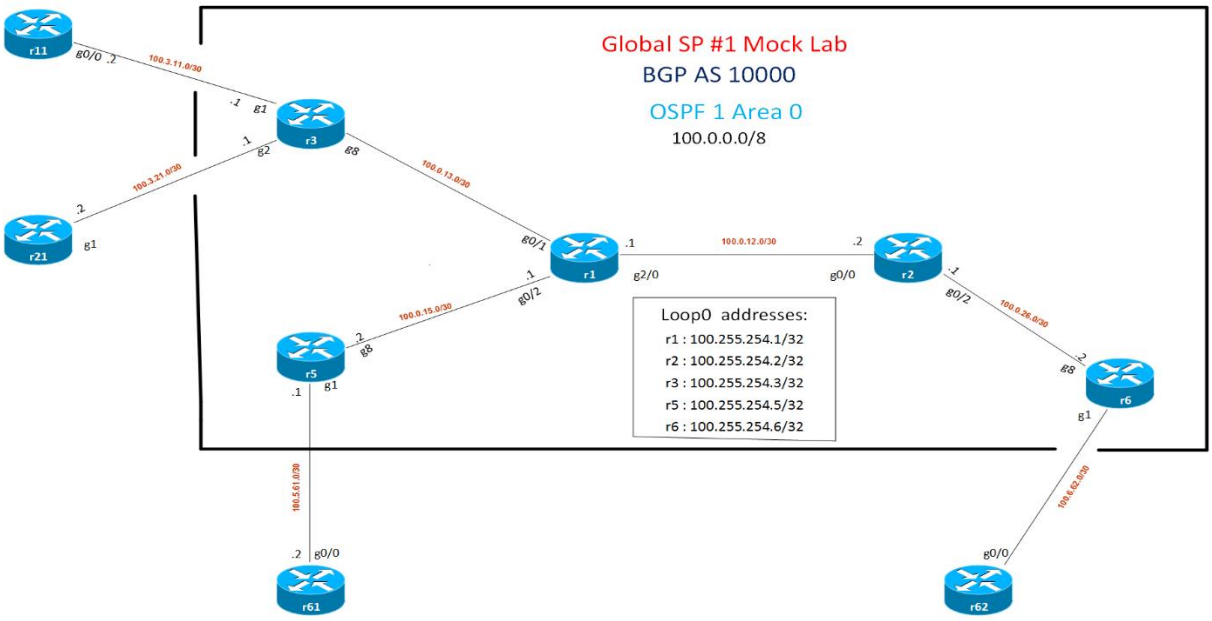
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FABD2 building SP #1 mock lab

Attempting to be an equal partner in technical discussions with Global SP #1. FABD2 has decided to build in mock lab of the MPLS closed run by Cinbal SP #1. Configuration will be tunnel on the Global SP #1 actual configurations of routers r5 through r5 while excluding router r4. To simulate the MPLS LJVPN locations. There will be three routers configured simarly to r21, r51 and r62. Because of budget constraints router r1 and r2 in the mock lab will also be shared with the FABD2 IT Training departments.

SP #1Topology



For a specific training scenario. The IT Training department already requested adding 200 additional loopback interfaces with unique IP addresses to routers r1 and r2 (100 loopbacks on each router) and advertising them in OSPF

One of them reasons for building MPLS mock lab is to gain a better understanding of how MPLS works, especially the label part. For the purposes of simplicity and clarity , users of the MPLS mock lab prefer to avoid seeing any prefixes, labels or any other runtime data that are not necessary for simulating an MPLS LJVPN with attached CE routers

Routers roles

- PE routers r3, r5, r6
- P router r3, r2
- CE router r21, r61, r62

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LDP Label Bindings

Avona	Travis
Travis with those 200 prefixes from IT Training Department, the label bindings on the routers in our MPLS lab are a mess. I'd like to filter out all unnecessary prefixes from LDP and keep only those needed for the MPLS LJVPN to work – so double-checking with you to avoid screwing something up Now, since Loopback0 is being used as the MPLS LDP RouterID in all the Ps and PE's, that loopback surely must be advertised with a label.	
	I am not 100% sure about that... Either way, we need labels for the IP addresses we use for our BGP VPNv4 peering – and there, we use Loopback 0 addresses, too. So the labels for those loopbacks need to be advertised. Anyway. But I don't think that we need any additional labels in LDP besides those.
Yeah… but you know…. The more I think about it. The more. This convinced that we must also have labels for the infrastructure links between the P's and the PE/PS – because the labels change hop by hop for the outer label. So I think the only prefixes we can filter out from LDP to avoid unnecessary labels are the 200 prefixes from the IT Training Department.	
	You may have point here…. This is getting tricky. Let's see what our CCIE candidate has to say.

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QUESTION 18

Refer to the new resource(s) available.

What mechanism and type of deployment would be the most appropriate to accomplish the label filtering goals as requested?

a) OSPF Prefix Suppression enabled globally on PE and P routers
b) OSPF Prefix Suppression enabled on the IT Training Departments 200 loopback interfaces
c) OSPF Prefix Suppression enabled on the links between PE and P routers
d) LDP advertisement filter applied to P routers
e) LDP advertisement filter applied to PE and P routers

Solution

Answer:

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Resources

LDP filtering too strict

From: Avona Vangyl"<.com>
To:"CCIE candidate"<.com>
Subject: LDP filtering too strict

Hey,

We've reviewed our discussion about the LDP filtering, and we'd like to loosen the rules up somewhat because the strict filtering makes us quite prone to missing some important networks we might have overlooked it's important to the that we do, we don't break the IP connectivity. How about allowing all labels except the training department's 200 loopbacks? Would that work? What would be the best way to implement that kind of filtering?

Avona
FABD2 RP/WAN team

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QUESTION 19

Refer to the new resource(s) available.

What is the proper approach to prevent the MPLS cloud from revealing its internal infrastructure to the attached endpoints?

a) Egress ACIs placed on PE-CE links
b) MPLS TTL Propagation disabled on PE routers
c) MPLS TTL Propagation disabled on routers
d) ICMP Unreacheables disabled on the Null0 interface on PE and P routers

Solution

Answer:

Resources

Protecting MPLS lab internals

From: Avona Vangyl"<.com>
To:"CCIE candidate"<.com>
Subject: Protecting MPLS lab internals

Hey,

I was testing the connectivity through our MPLS lab, and things work nicely. But one thing I don't like is the fact that the MPLS lab reveals its internal addressing when doing traceroute, for example. Our real provider does not do that. How are they doing that? ACLs? I'd like to have their behavior replicated as closely as possible.

Travis
FABD2 Network Manager

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QUESTION 20

Refer to the new resource(s) available.

Given the description of the issue, which of the following statements would explain the symptoms described in the e-mail from Travis?

a) The hosts resolved their own hostnames to IPv6 addresses in DNS
b) IPv6 unicast routing was not enabled on sw101
c) The M-flag was not set in Router Advertisements
d) There was no IPv6 IGP running in VLAN 2001

Solution

Answer:

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Resources

IPv6 implementation in HQ

From: Avona Vangyl"<.com>
To: "CCIE candidate"<.com>
Subject: IPv6 implementation in HQ

Hey,

We've been considering deploying IPv6 in our HQ in a limited fashion, just one VLAN for the time being, to play with it. To make things simple, we've configured sw101 as the DHCPv4 server for VLAN 2001, enabled IPV6 on the SVI assigned the global prefix to the SVI, and at first, it seemed like everything was working okay – but then we realized that there are no bindings created on the DHCPv6 server, and what's more. When we removed the DHCPv6 configuration from sw101 (so part of our preparations to migrate it to the central DHCP server as sw211.) the hosts in the VLAN 2001 were still getting IPv6 addresses with the proper profile. But how's that possible? There's no static IPv6 configuration on the end hosts anywhere

Travis
FABD2 Network Manager

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QUESTION 21

Refer to the new resource(s) available.

Given the description of the issue, what are the two reasons for the absence of RAs breaking the IPv6 connectivity? (Choose two.)

a) The end hosts considered the IPv6 to be disabled in their network.
b) The end hosts could not locate their default gateway.
c) The sw101 and sw102 switches stopped routing IPv6 traffic on SVI for VLAN 2001.
d) The sw101 and sw102 switches stopped advertising the global prefix on SVI for VLAN 2001 in EIGRP
e) The end hosts could not locate their DHCPv6 server
f) The end hosts did not have the necessary information for an autoconfiguration mechanism

Solution

Answer:

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Resources

RE: IPv6 implementation on HQ

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: RE: IPv6 implementation on HQ

Hello,

Thanks for the response! We have fixed the configuration and DHCPv6 started working

We have started extending the IPv6 further in the HQ, aside from sw101 and sw102 VLAN 2001, we also extended it southward to r11 and r12 – we just enabled IPv6 with link-local addresses on those connections and started running EIGRP for IPv6 across all those devices, and we could see r11 and r12 learning the global prefix from sw101 and sw102 nicely. Then, following some best practices we read about, we decided to lighten down our IPv6 environment in VLAN 2001, and so we disabled RAs on sw101 and sw102 SVIs for VLAN 2001 to prevent leaking address information and having hosts jump on that intend of using DHCPv6, But this broke our IPv6 connectivity in VLAN 2001, so we had to receivable them. But I wonder – why this happen?

Travis
FABD2 Network Manager

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QUESTION 22

Refer to the new resource(s) available.

What would be the proper approach to meet the security requirement as stated by Travis?

a) Implement IPv6 Secure Neighbor Discovery (SeND)
b) Enable RA Guard
c) Suppress the prefix information in RAs
d) Decrease the frequency of sending out RAs

Solution

Answer:

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Resources

RE: RE: IPv6 implementation on HQ

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: RE: RE: IPv6 implementation on HQ

Hey,

Okay, so we renabled RAs. But how can we have the cake and eat it, too? We definitely want to avoid some wrongly implemented hosts to continue ignoring DHCPv6 and inserting their own addresses, even if they fall into the proper global prefix scope, and overall, we don't want some eavesdropper to given on the sensitive information from the RAs.

Ideas?

Travis
FABD2 Network Manager

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QUESTION 23

Refer to the new resource(s) available.

23.1 This item consists of multiple questions you may need to scroll down to be able to see all questions

For each gateway redundancy mechanism, select which characteristics are applicable on an IOS-based platforms, if any (select all that apply)

	HSRP	VRRP	IPv6 RA
Active role in one instance can control roles in other instances			
Non proprietary mechanism			
Active role can be coupled with mechanisms such as DHCP Relay or IPsec			
Support active-active load balancing out of the box			
Transparent to end hosts			
Can be coupled with BFD			

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Solution

	HSRP	VRRP	IPv6 RA
Active role in one instance can control roles in other instances			
Non proprietary mechanism			
Active role can be coupled with mechanisms such as DHCP Relay or IPsec			
Support active-active load balancing out of the box			
Transparent to end hosts			
Can be coupled with BFD			

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23.2 Given Travis preference, what would be the first hop redundancy mechanism of choice?

a) HSRP or VRRP
b) VRRP or IPv6 RAs
c) HSRP only
d) VRRP only
e) IPv6 RAs only
f) HSRP or IPv6 RAs

Solution

Answer:

Resources

RE: RE: RE: IPv6 implementation on HQ

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: RE: RE: RE: IPv6 implementation on HQ

Hi,

As part of implementing IPv6 in HQ, we have been thinking about IPv6 gateway redundancy, too. We've been looking at HSRP, VRRP AND IPv6 RAs. They all seems to essentially provide the same level of redundancy is fact, the difference between them are kind of blurry to me. Can I tap into your expertise once again to better for this?

In the end, we would like to pick and tune the protocol to allow end hosts to switch over to another gateway within 1 second is most, does not inundate the end hosts with excess traffic to process, and whole inner workings are simple.

One thing – I definitely prefer open protocols. I will only go with proprietary mechanisms if I have no other choice.

Cheers

Travis
FABD2 Network Manager

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QUESTION 24

Refer to the new resource(s) available.

When building the overall SD-WAN policy to meet the Payment Card Industry requirements for the Point Of Sale (POS) terminals at Branch #1 and Branch #2, what three steps must be accomplished in vManager? (Choose three.)

a) Create an ACI at Branch #1 and Branch #2 blocking their direct mutual communication
b) Create POS VPN AND VPN interface feature templates and apply them to Branch #1 and Branch #2 device templates
c) Apply the policy outbound to the Site IDs of Branch #1 and Branch #2
d) Apply the policy outbound to the Site ID of the DC
e) Create a policy to set the TLOCs for Branch #1 and Branch #2 POS OMP routers to the DC TLOC(s)
f) Block Branch #1 and Branch #2 from learning each other's TLOC routers

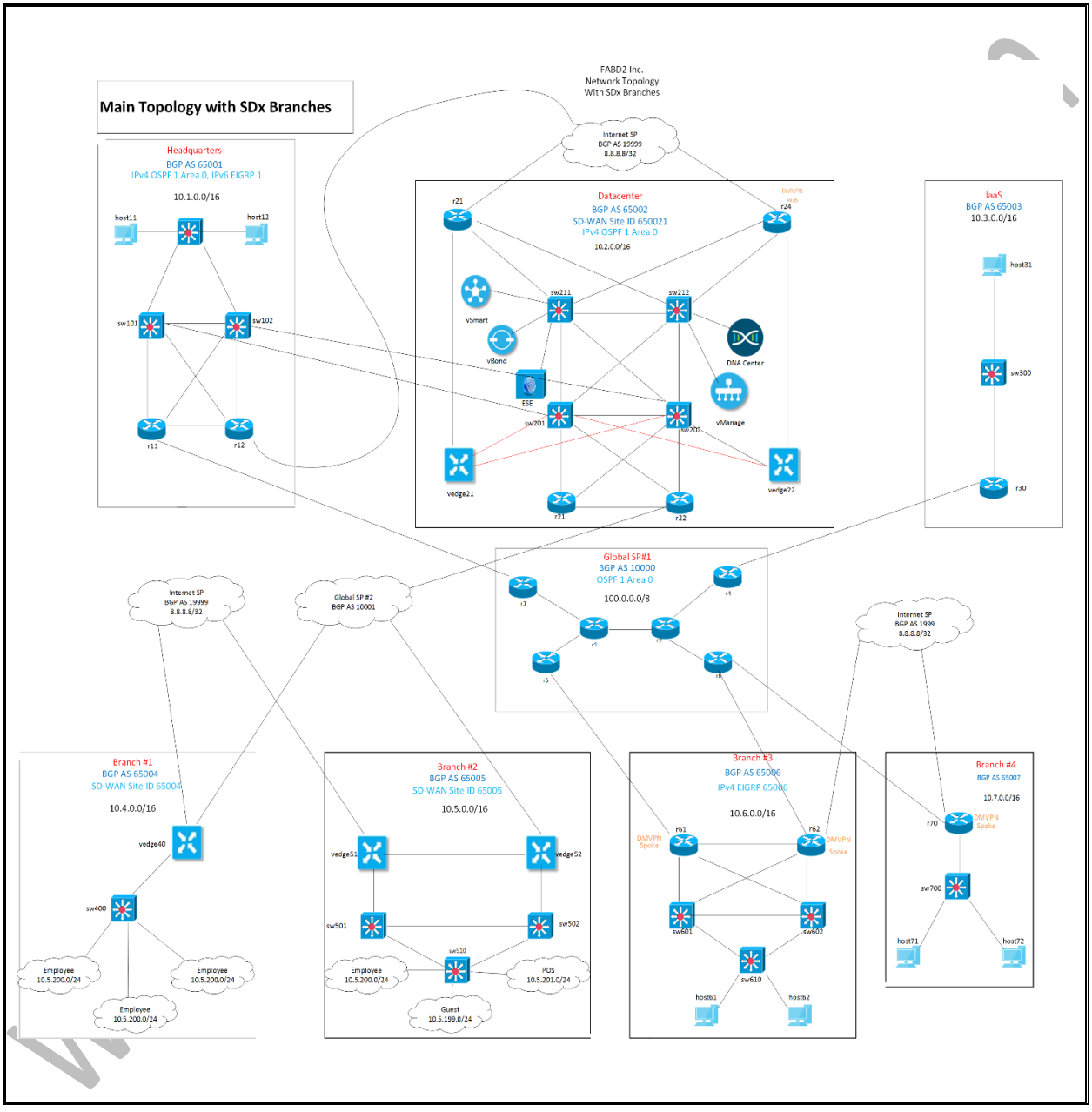
Solution

Answer:

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Resources



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SD-WAN PoC

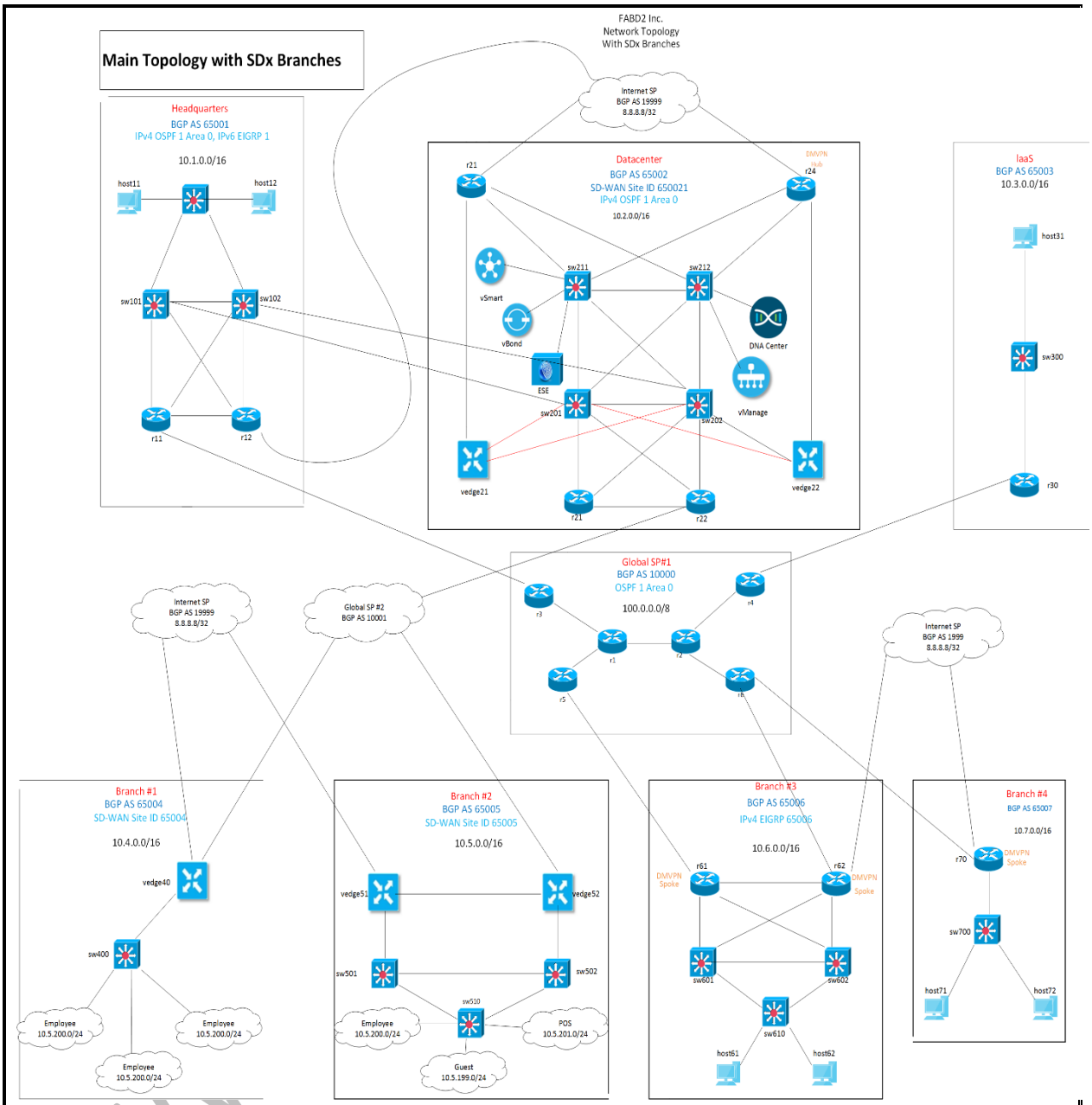
From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: SD-WAN PoC

Hi,

While we have been keeping you busy in other thing, we have already started yet another project in the background. A Proof of Concept implementation of SD-WAN and SDA on two new sales. Branch #1 and Branch #2. These branches are connected through in different service provider, Global SP #2, which in terms connects them to the router r22 in DC. Also, both branches connect to our ISP.

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Since SD-WAN deployment has already been done, on both Branch #1 and Branch #2, we have created two VPNs. Employee and Guest, and these are working in the full mesh mode just fine. Now, however, we need to extend both the branches and DC with another VPN for Port Of Sale (POS) terminals. Since these terminals process credit cards, it is imperative that the Payment Card industry (PO) requirements are not. In short, these are requirement

- On each branch, Port Of Sale (POS) terminals must be on a different network segment, isolated from any other networks on the branch.

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- Under no circumstances may POS terminals on Branch #1 communicate directly with POS terminals on Branch #2 and vice versus. Any such communication be instead routed through the data center where we have the necessary firewalls in place.

This is departure from the full mesh SD-WAN we have right now, and I am not entirely certain how to implement it. I'd appreciate your guidance here.

Travis
FABD2 Network Manager

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QUESTION 25

Refer to the new resource(s) available.

Based on the given constraints and existing design, which two steps can be performed to provide WAN transport redundancy at Branch #2 (Choose two.)

a) On the link between vedge51 and vedge52, create 802.1Q subinterfaces as necessary and use them as TLOC extensions for each vEdge's transport
b) Add a second physical link between vedge51 and vedge52 and use the links as TLOC extensions for each extensions for each vEdge's transport
c) Configure a backup default route on each vEdge pointing to the address of the neighboring vEdge's TLOC extension interface
d) Configure an outbound localized policy on each vEdge to add the TLOC of the neighboring vEdge to the advertised OMP routes
e) Run OMP between vedge51 and vedge52

Solution

Answer:

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Resources

RE: SD-WAN PoC

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: RE: SD-WAN PoC

Hello

I've got some doubts Branch #2 and its WAN redundancy

When we originally designed Branch #2, we wanted to purchase two MPLS circuits and two internet circuits to ensure transport independence. But now the management started having second thoughts – they said they won't sanction additional expenses. That means that we are stuck with what have right now – we cannot procure additional equipment, install additional physical connections, circuit, or use additional providers. The management who said that since both Branch #1 and Branch #2 have a direct connection to the ISP, we should avoid backhauling the element bound traffic from these branches through the data center.

I was wondering it, given our current design we could still somehow leverage our redundant connections at Branch #2

Travis
FABD2 Network Manager

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QUESTION 26

Based on the given constraints and existing design, which two steps can be performed to ensure that internet-bound traffic from Branch #2 is not sent via the data center?(Choose two.)

a) On Vedge52, configure NAT to VPN 0 on the interface connected to the vedge51 TLOC extension interface for the internet transport.
b) On vedge51, configure NAT to VPN 512 on the interfaces toward the ISP.
c) On vedge51, configure NAT to VPN 0 on the interface toward the ISP.
d) On vedge52, configure NAT to VPN 0 on the interface toward SP #2.
e) On vedge51, configure NAT to VPN 0 on the TLOC extension interface for the internet transport.

Solution

Answer:

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QUESTION 27

Refer to the new resource(s) available.

Which two steps are required to implement the desired Guest VPN design? (Choose two)

a) Implement a localized data policy that blocks Guest VPN traffic between SD-WAN branches.
b) Configure a centralized VPN membership policy that only allows Guest VPN prefix to be advertised in OMP.
c) Configure a centralized VPN membership policy that restricts the Guest VPN prefix from being advertised in OMP.
d) Configure centralized data policy that perform NAT of Guest VPN traffic to VPN 0.
e) Configure a localized control policy that rewrites the TLOC of Guest VPN routes in OMP to 0.0.0.0.

Solution

Answer:

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Resources

Guest VPN addressing on branches

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: Guest VPN addressing on branches

Hello,

Thank you very much for the help with the SD-WAN technology so far. Really amazing work!

One more request… We have decided to streamline our Guest VPN deployment on every current and future SD-WAN enabled branch. Instead of allocating unique IP prefix for the Guest VPN on a per branch back, and given the fact that we don't provide any services aside from internet access in the Guest VPN, we have decided to use 10.100.100.0/24 as the Guest VPN prefix at every SD-WAN enabled branch . You already helped us to make sure that internet-bound traffic will break our locally, and this remains valid for the Guest VPN even after this change. In addition, we also want to ensure that there is no connectivity between Guard VPNs across branches.

Can you suggest the steps we need to know? Thanks!

Travis
FABD2 Network Manager

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QUESTION 28

Refer to the new resource(s) available.

Given the intended scope of SDA fabric deployment on Branch #2, which option represent the smallest applicable IP pool in DNA Center to support LAN Automation on Branch #2?

a) one /24 subnet
b) one /26 subnet
c) one /27 subnet
d) two /26 subnet
e) one /25 subnet

Solution

Answer:

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Resources

SDA addressing

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: SDA addressing

Greetings,

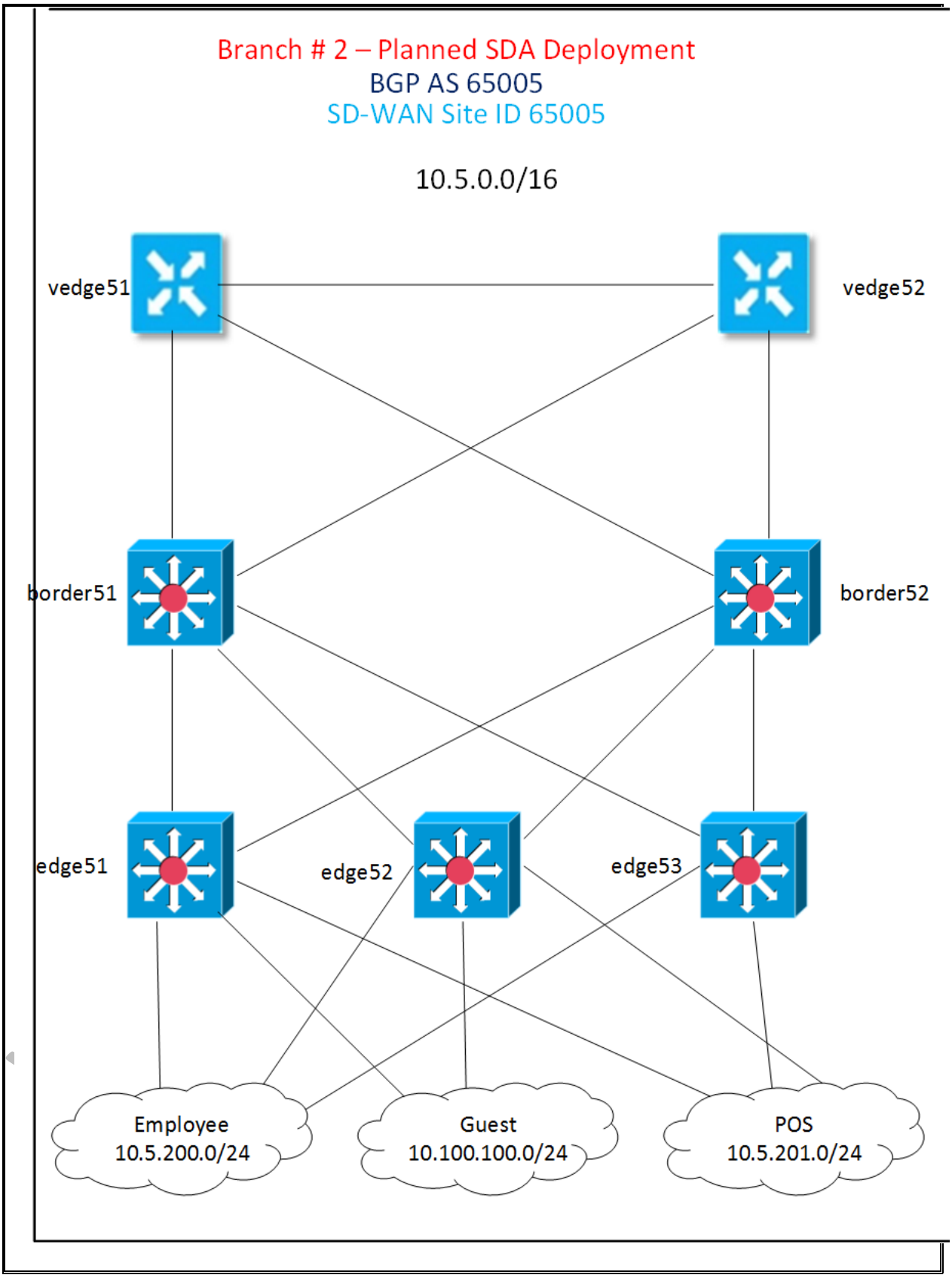
Now that we have the SD-WAN topology and associated policies successfully in place. I am looking at utilizing SDA fabric in Branch #2.

We've discussing something that is not deployed on the branch yet, and we're not sure if we are even going to implement this, but we wanted to know some ideas around and discuss them with you.

From what we know. DNA Center can provision the underlay itself – it's called the LAN Automation. However, there seems to be certain requirements regarding the address pools put into DNA Center so that the LAN Automation works. The SDA fabric at Branch #2 may consist of up to three edge and two border switches, with each edge switch having a connection to both border switches. Once again, the SDA is not implemented on Branch #2 yet but it is planned. Here's a diagram of the planned deployment.

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QUESTION 29

Refer to the new resource(s) available.

Which option represents the smallest applicable IP pool in DNA Center to support the planned Layer3 VN handoffs on Branch #2?

a) one /25 subnet
b) one /26 subnet
c) one /24 subnet
d) two /26 subnet

Solution

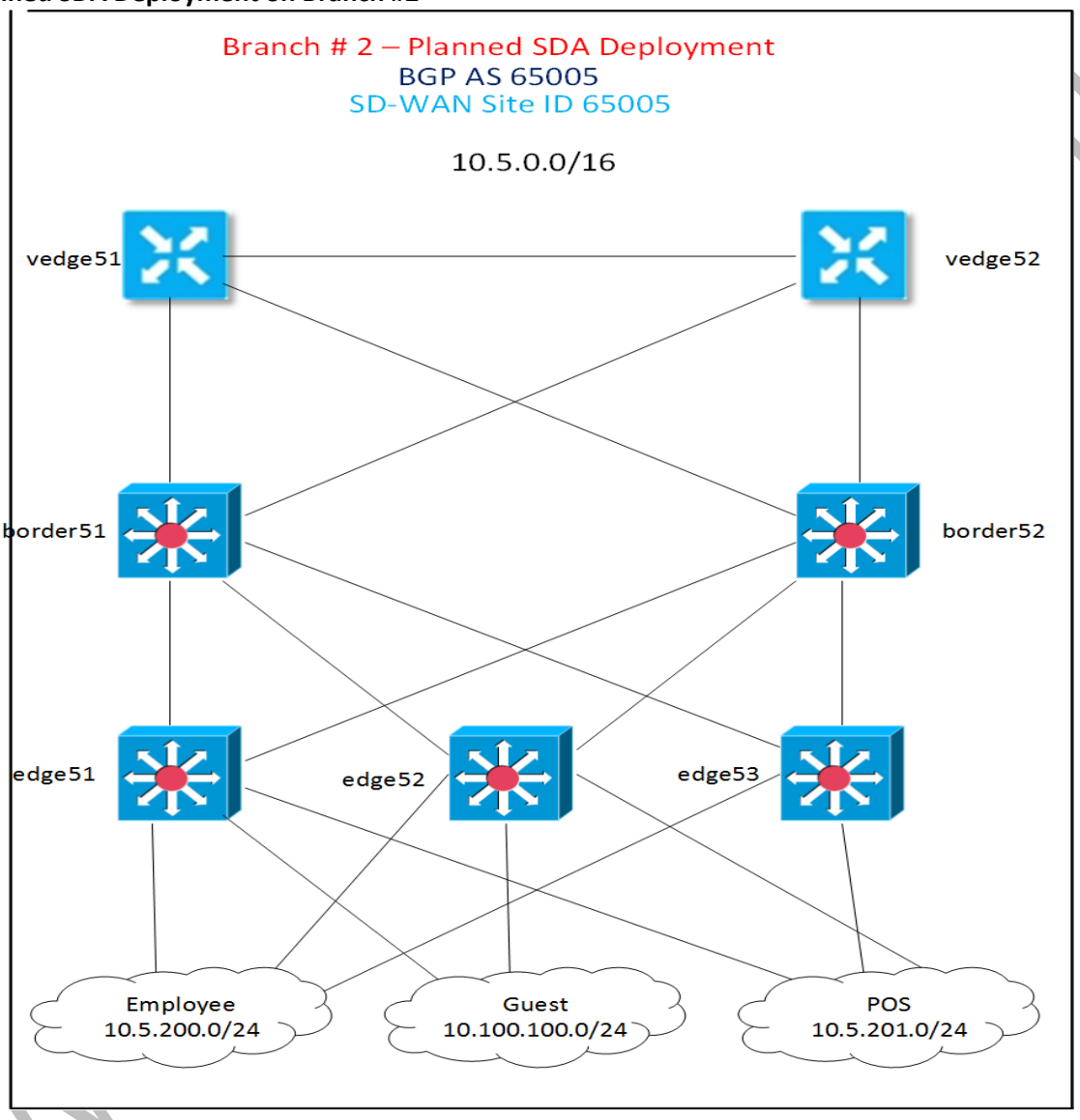
Answer:

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Resources

Planned SDA Deployment on Branch #2



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RE: SDA addressing

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: RE: SDA addressing

Hello,

I totally forgot to ask you.

We will be deploying three VNs on Branch #2 Employee, POS and Guest. We would like DNA Center to automotive that L3 handoff from the border nodes to the vEdge routers as well, along with DEFAULT_VN and INFRA. What should be the smallest IP pool in DNA Cente we need to allocate for these handoffs?

Thanks!

Travis
FABD2 Network Manager

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QUESTION 30

Refer to the new resource(s) available.

Which two design options are applicable to provide transit between planned SDA fabrics in Branch #1 and #2, considering the future plans? (Choose two)

a) Deploy IP Transit between Branch #1 and Branch #2
b) Deploy a Transit Control Plane node in Data Center to facilitate the transit between Branch #1 and Branch #2
c) Deploy SDA Transit between Branch #1 and Branch #2
d) Use BGP as a handover protocol between SDA border nodes and SD-WAN vEdge routers
e) Combine Branch #1 and Branch #2 into a single multi-location SDA fabric site

Solution

Answer:

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CCIE Lab Center

Resources

Transit between SDA branches

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: Transit between SDA branches

Hello,

The management come back with a question. Assuring we had Branch #2 already running on SDA, and we also implemented SDA on Branch #1 with the same VNs, what would be the way of interconnecting these branches when a single fabric doesn't? The SD-WAN as a transport technology stays, of course, but the SDA documentation takes about two types of interconnections. IP Teams and SDA Teams. The SDA Teams seems to be far superior – allows carrying SGTs end-to-end is able to extend VXLANs if necessary. But I am not sure if that is possible with SD-WAN setting between the branches, on top of it, we are looking at leveraging Application Aware Routing in SD-WAN in future for the transisitng VNSVPNs.

So I'm unsure what options we have here. Can I pick your brain once again?

Travis
FABD2 Network Manager

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QUESTION 31

Refer to the new resource(s) available.

Drag the options on the left and drop them in any order into the two corresponding categories on the right, indicating the best practice where these options should be added in DNA Center. Not all options are used

	DNA Center GUI Workflow	DNA Center Template
UDLD	Option 1	Option 1
Anycast GWs	Option 2	Option 2
VTY ACLs	Option 3	Option3
Spanning Tree (MST)	Option 4	
SNMPv3		
TACACS+servers		
Port Security		
Application Policy		

Solution

	DNA Center GUI Workflow	DNA Center Template

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Resources

DNAC is… confusing

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: DNAC is… confusing

Hi,

The DNA Center appliance in our data center is now up and running. I'm trying to start out by building a few policies, but I am a bit confused, since this is all new territory for me. (And probably the rest of us too.) I have analyzed the configuration is the existing infrastructure and picked a few areas that I want to build out in DNA Center but need some help getting started. The areas the working on are:

1. TACACS+ servers
2. UDLD
3. Spanning Tree (MST)
4. Port Security
5. SNMPv3
6. VTY ACLs
7. Anycast GW
8. Application Policy

Can you help me out?

Travis
FABD2 Network Manager

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QUESTION 32

What are two possible ways of ensuring that authorized local administrators in the Employee VN on Branch #1 or Branch #2 can still access the local SDA border nodes using their loopback addresses through in-band SSH access? (Choose two.)

a) Utilize an external firewall for controlled inter-VN communication.
b) Utilize a vEdge router as a fusion router.
c) Deploy console terminal servers.
d) Implement IS-IS redistribution between VNs.
e) Set up fabric SGACLs permitting this communication.

Solution

Answer:

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QUESTION 33

Refer to the new resource(s) available.

What are the two valid design options for deploying QoS on the SDA branches that will meet FABD2 requirement? (Choose two.)

a) Extend the existing queuing model into a new 4/5 class model.
b) Use the DNA Center templates to rebuild the QoS policy.
c) Leverage the SGT-based QoS.
d) Use the DNA Center to define business-irrelevant application sets.
e) Use the DNA Center application policy to rebuild the QoS policy.

Solution

Answer:

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Resources

QoS policies for SDA

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: QoS policies for SDA

Hi,

Since forever, we have been using switch-based QoS policies as we haven't had a tool to deal with QoS provisioning at all. In our legacy deployment, our ingress QoS policy is to treat DSCP makings, and on agress, we are currently using the 1 priority queue, 3 round robin queue, 8 weighted drop thresholds queuing setup.

The approach has always been causing some headaches. Some of the most commonly heard issues or complants. I have received are:

- Not able to accommodate new video applications or meet future growth.
- Can only match applications based on L3 and L4 information, not based on true applications flows
- Not able to mark down irrelevant to our business
- Hard to maintain consistent QoS policy acess different platforms

Now that we are looking into implementing QoS in our Branch #1 and Branch #2, we would like to make it right, adopting the Cisco LAN QoS best practors and avoiding the issues I have listed above I am reaching out to you with a request to provide some input as to how we should approach this in an SDA content.

As always, thanks!

Travis
FABD2 Network Manager

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QUESTION 34

Refer to the new resource(s) available.

Given the requirement, what would be the best way to implement the logging on r21?

a) SNMP poling and processing the results offline
b) Local scripting on the router using a procedural language
c) NETCONF poling and storing results on the routers
d) Use a Python script to access the router CLI remotely through SSH and drive the output collection

Solution

Answer:

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Resources

Saving command outputs

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: Saving command outputs

Hello,

We have recently been having issues with the SP #1 on r21 in DC – I'm really glad that we'll be moving off to SP #2, but till then, I have to put on a fight with them to have the stuff fixed, and need some evidence is test them over their heads with

What's happening is this from time to time, the BGP session between r21 and the SP#1 PE router flaps. When it comes up again, it takes several minutes before we learn any prefixes from the PEs SP#1 in taking us that we are not aware of any issues with their PE's but I am suspecting that they are somehow late in scheduling the updates to be sent to us. What I would like to do in a logging setup that would record these outputs.

- Snapshot of the BGP neighbor statistics (show ip bgp summary)
- Complete BGP (show ip bgp)
- Every received route from the respective PE pulled out through show ip bgp neighbor X.X.X.X routers. Its detailed view (show ip bgp N.N.N.NM)

I want the logging to be done automatically and independtly by r21. Both periodically every minute, and on top of it, in the moment of the BGP session coming up. I intend to allow SP #1 a read only access to r21 s)

I would appreciate any thoughts on how to implement this.

Travis
FABD2 Network Manager

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QUESTION 35

Refer to the new resource(s) available.

Which are the characteristics of the different scripting method? (For every scripting method, select all characteristics that apply.)

	EEM Python Policy	EEM Applet calling a standard Python script	Standard Python script without EEM
Requires guest shell			
Allows sharing the same Python script for periodic and triggered collection			
Allows scheduling a periodic collection run			
Allows triggering the collection run on a BGP session event			
Allows running the Python script manually outside EEM			

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Solution

	EEM Python Policy	EEM Applet calling a standard Python script	Standard Python script without EEM
Requires guest shell			
Allows sharing the same Python script for periodic and triggered collection			
Allows scheduling a periodic collection run			
Allows triggering the collection run on a BGP session event			
Allows running the Python script manually outside EEM			

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Resources

RE: Saving command outputs

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: Saving command outputs

Hi,

Thanks for the ideas. Taking them into account, and reversing the available IOS-XE docs. I've decided to proceed with local Python scripting on r21.

I have to say though that the docs are not entirely clear to me. I initially thought that I need to write a Python script that does everything itself – run is an infinite loop, waiting for either the 1-minute timer is expire or for the EGP – session event to line, do the collection as intended, and keep this repeated. But the documentation seems to suggest that I need to use EEM and from these, call the Python script. They ever talk about on EEM Python Policy using the 'Import ever'
Which I don't understand how it differs from a normal Python script.

Can you help me understand what the differences here are?

As always – appreciated

Travis
FABD2 Network Manager

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QUESTION 36

Refer to the new resource(s) available.

Given the circumstances, what is the best option for Anna to develop and debug her scripts before deploying them on FABD2 production network?

a) Use the production network while executing REST API calls bundled in a transaction and rolled back at the end without a commit
b) Perform the development and debugging on the production network during dedicated maintenance windows
c) Create a lab repro for development purposes
d) Use DevNet SD-WAN sandbox labs

Solution

Answer:

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Resources

SD-WAN Automation Development

Avona	Travis
Travis, I would like to test out a couple of automation scripts for our SD-WAN deployment I've been developing myself – was looking into gathering data such as inventory of devices, some real-time monitoring and remote device reboots. I am so excited to put this testing done! Would love to start right away. Just wanted to list you know ☺	
	Wait. You want to test the scripts you're developing yourself on our networks?
Yep ☺ Don't worry. Those APIs are non-disruptive	
	Avona. I don't think it's a good idea. This is a production network, you know I trust you, but I don't want experiments on it at any point in time. The APIs may be 'non-disruptive' in themselves, but a minor mistake in the script could render even a harmless, API call disruptive. Just call it in a tight infinite loop, and we have a DoS on our hands. So, sorry. But no. No development and testing on our production network, only something that has been tested and debugged elsewhere.
Well.. Oh ☹ But what are my options, then? Nobody writes flawless code without need to debug it. Every now and then. Should I do a lab repro of our SD-WAN, then? But that'll take lots of time to set everything up, and we'd need to buy extra license for it.	
	Hmmm. Lab repro is an option, but requires considerable investments, which is outright impossible to justify in this case. But let's ask our CCIE –to-be what options we have for doing this kind of testing without putting our production network in jeopardy

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QUESTION 37

Refer to the new resource(s) available.

This item consists of multiple questions you may need to scroll down to be able to see all questions

37.1 What authentication mechanism is used for API calls to vManage?

a) basic HTTP authentication with every API call
b) authentication token in HTTP headers obtained after a call to/auth/token with credentials passed as HTTP basic authentication
c) client X 509 PKI certificate presented with every API call
d) session cookies obtained after a call to /I_security_check with credentials passed in the request body

Solution

Answer:

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37.2 What is the nature of the value for the deviceId key for a vEdge?

a) hostname
b) license number
c) device channels number
d) certificate serial number

Solution

Answer:

37.3 What is the purpose of enclosing the deviceIP / deviceId object into square brackets in the JSON call template?

a) The request can hold multiple deviceIP / deviceId object as a list
b) The square brackets and readability but are not mandatory
c) The square brackets introduce an optional part of the request
d) The deviceIP / deviceId object is a nested object inside another one, with nesting requiring the use of square brackets

Solution

Answer:

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Resources

Questions regarding SD-WAN API calls

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: Questions regarding SD-WAN API calls

Hi,

So, I've started playing with the SD-WAN APIs but I realized that I was too optimistic about me understanding how they work. Can I pick your train on this?

For my experiments, I have picked the vEdge reboot API call. The URL should be https://vmanage/dataservice/device-action/reboot and the JSON request template from the docs looks like this

(
"action" I "reboot"
"deviceType" I "vedge"
"devices" : |
(
"deviceIP" I "1/2.16.255.11"
"deviceIP" I "face13fa.d dn1-4000-9344-s1000p04o14dd"
)
)
)

I have a couple of doubts here.

- How do we authenticate to vManage? I don't suppose the API is unprotected for anyone to just send API calls as they please, but the docs for the reboot API did not mention any authentication at all.
- This devideid in the template – I don't know what value to put there
- Why is the 'deviceIP' pair enclosed in the square [] brakets? They seems to simply double the curly [] brackets which I see point for

Thanks.

Avona
FABD2 RP/WAN team.

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QUESTION 38

Refer to the new resource(s) available.

Which two of the following changes to the script would shorten its running time without impairing its functionality? (Choose two.)

a) Construct the JSON body of the request manually instead of using the json.dumps0 method.
b) Execute the loginAPI0 only once and reuse the session for multiple API calls.
c) Use the put0 method instead of post0 to pass the reboot API call.
d) Combine device IP/ID pairs into a list and pass them all in a single API call.
e) Refer to the vManage by its DNS FQDN instead of its IP address.

Solution

Answer:

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Resources

RE: Questions regarding SD-WAN API calls

From: 'Travis Handerson'<.com>
To: 'CCIE candidate'<.com>
Subject: RE: Questions regarding SD-WAN API calls

Hi,

Thanks a lot! It worked!

I have one more question if you don't mind. I've been looking at the performance of my script, and to stand the API calls to reload 2 vEdges, it takes the script around 2 seconds to complete. I played around with the script, adding more vEdge IPs and IDs to find out how the amount of vEdge plays a role in the script execution time, and it seems that sending a reload request for any single vEdge takes about 1 second, the time growing linearly with the number of vEdge to reload. That feels way too long to be honest. Something's wrong here that's delaying the run of my script, but I cannot find out what. Can you have a look at it? Oh, I've omitted the body of the loginAPI() function to keep the below output short.

```
Import request
Import sys
Import json
From requests.packages.erilliso.exceptio.import.Insecurerequestwarning

Requests.packages.er 1lip3.fixavle_warning (InsecureRequestWarning)

.def .loginAPI (host, username, password) :
* body omitted
  Return session


Def reloadAPI (host, username, password, ip, id)
 session * LoginAPI (host, username,password)
 request * ('action' :  'reboot' , 'deviceType' : 'vedge' , 'devices' I [] }
 request | 'device' | append  (  {'deviceIP' : IP , 'deviceId' : id ) }
 sp1 – 'https://'' * host – '/dataservice/device/action/reboot'
 headers * ( 'Content - Type'  | 'application'json' )
 return session.port (SP1 SPI 1, data*json.dumps (request), headers header, verifyfalse)

device * {}
devices update (  ( '1.1.2.3'  :  'o7c150fb-1tbe-2tb4-2be1-2sbdcb9?afc3'  )  )
```

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devices update (('1.1.2.3' : '2e24c1de-7tbe-2235-abc9-93492abccdb'))

For ip in devices
Response : reloadAPI ('10.2.263.11'. 'admin', 'admin' , ip , devices {ip}))
Print (response-content)

Thanks!

Avona
FABD2 RP/WAN team

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QUESTION 39

You have reached the end of exam module 1.

Click "End Exam Section" in the main screen in order to proceed to module 2.

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**********The End**********

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